

Chapter 3

NMR properties of fluids

3.1 Introduction

In designing a down-hole NMR logging tool in the 1950s, the researchers initially assumed that the hydrogen nuclei in oil and brine saturated rocks would relax at their bulk rates. The differences in relaxation rate would then be used to identify the oil and brine fractions in the rocks. Because the amplitude of NMR relaxation measurements was proportional to the amount of hydrogen sensed, they presumed also that the tool would be able to determine formation porosity.

Subsequent studies showed that the NMR relaxation of brine in rocks is heavily dominated by surface relaxation. Unrelated studies also showed that the T_2 signal was subject to an additional relaxation enhancement due to diffusion of the nuclei in a magnetic field gradient. Thus, to interpret relaxation measurements in fluid saturated rocks, we need to sort out the contributions from bulk and surface relaxation as well as the diffusion effect. We shall start with NMR relaxation measurements on the bulk fluids typically found in hydrocarbon reservoirs in this chapter, and then cover surface relaxation and diffusion effects of these fluids in porous media in Chapter 4.

Saturating Fluids

In NMR studies of porous media for petrophysics, the saturating fluids of interest are water (of variable salinity), gas, and oil. The water salinity can range from nearly pure to salt saturated (most commonly the salt is NaCl). The gas we discuss is the free gas, mainly methane, i.e., in gaseous form,

rather than dissolved in oils. The oil can be free of gas (dead oil) or gas saturated (live oil). Live oils exist at elevated temperatures and pressures where the gas dissolved in the oil is below its bubble point. Synthetic oils used in drilling muds can invade the reservoir either displacing or mixing with the reservoir fluids, and are thus also of interest to the oil industry.

Hydrogen Index

Using NMR logging tools to determine the porosity of a hydrocarbon reservoir is based on calibrations of the tools with respect to known volumes of water. Such calibrations require that the NMR signal per unit volume of each saturating fluid be known. Thus it is convenient to introduce the term hydrogen index (HI) to quantify the atomic hydrogen densities of oil or gas relative to that of an equal volume of water. The hydrogen index is defined as the amount of hydrogen per unit volume of a fluid divided by the amount of hydrogen per unit volume of pure water in standard conditions, i.e.,

$$\text{HI} = \frac{\text{Amount of Hydrogen in Sample}}{\text{Amount of Hydrogen in an equal volume of Pure Water}}. \quad (3.1)$$

Thus hydrogen index should be a quantity independent of measurement methods. Sometimes, when there are limitations in measurement methods, *apparent* rather than *true* hydrogen index is measured.

Oil field brines exhibit a wide variation in salinity. At low salinity, the HI of brine is approximately 1, but at high salinity, the HI can be reduced by as much as 10% (see Figure 3.1), as a small but significant fraction of the brine volume is taken up by the dissolved salt.

The hydrogen indices of most “dead oils” are very close to 1, but the hydrogen index of gas is strongly dependent on its pressure. When the density (ρ), the number of hydrogen atoms in the chemical formula (n_H), and the molecular weight (M_W) of the oil or gas are known, the following equation can be used to calculate HI at room conditions

$$\text{HI} = \frac{9\rho n_H}{M_W}, \quad (3.2)$$

where the number 9 is the ratio of the molecular weight of water (18 g) to the product of the number of hydrogen atoms in a water molecule (2) and the density of water (1 g/cm³).

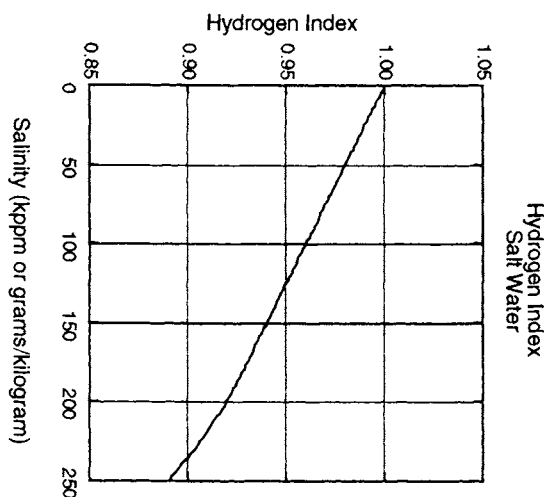


Figure 3.1: Hydrogen index of salt water. (From Kleinberg and Vinegar [128].)

For a light alkane such as decane, the calculation is as follows. Decane has the formula $C_{10}H_{22}$. Thus $n_H=22$ and the molecular weight is 142. The measured density of decane is 0.7287 g/cm^3 . Substituting these numbers in the HI calculation (Eq. (3.2)) yields $HI=1.016$. Similarly, hexane (C_6H_{14}) with a density of 0.6597 g/cm^3 and a molecular weight of 86 has $HI= 0.9665$.

Kleinberg and Vinegar [128] reported HI values for a suite of low and medium density oils. Their HI values are plotted versus API gravity in Figure 3.2. For the low density oils (high API gravity), the HI was approximately 1. However, they showed a decrease in HI as the API gravity decreased below 20. They attributed at least some of the decrease in HI to insufficient sampling of the short T_1 and T_2 values of the heavier oils, thus the measured HI may be the apparent HI rather than the true HI.

Note that API gravity¹ of a crude oil is related to its density ρ (in gm/cm^3)

¹The density of crude oils ranges from about 1.01 to 0.75 gm/cm^3 . Thus, the corresponding API gravity varies from about 8 to 57.

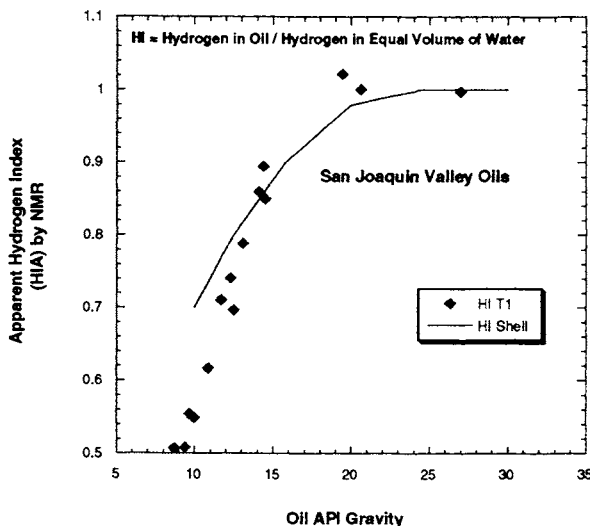


Figure 3.2: Hydrogen Index (HI) versus API gravity. (The solid line is from Kleinberg and Vinegar [128] and the data points are from LaTorraca et al. [137].)

through the following definition:

$$\text{API gravity} \equiv \frac{141.5}{\rho} - 131.5. \quad (3.3)$$

It is a scale (API or American Petroleum Institute scale) commonly used in the petroleum industry to characterize the quality of a crude oil.

LaTorraca et al. [134, 135] reported HI measurements on a suite of medium to high density oils. Their results are also shown in Figure 3.2. The viscosity values of these oils varied from ~ 1000 cp to $\sim 100,000$ cp at room temperature. Both the combustion and low field T_2 methods were used to determine the range in HI. From the combustion method, only the highest density crude oils, with viscosity values greater than $\sim 10,000$ cp, had HI values less than one as shown in Figure 3.3. Note that the NMR and combustion methods coincide only for the two oils with the lowest viscosity and density. The HI estimates are plotted versus viscosity because in certain well logging situations it is possible to estimate oil viscosity from the difference between NMR and density porosity determinations.

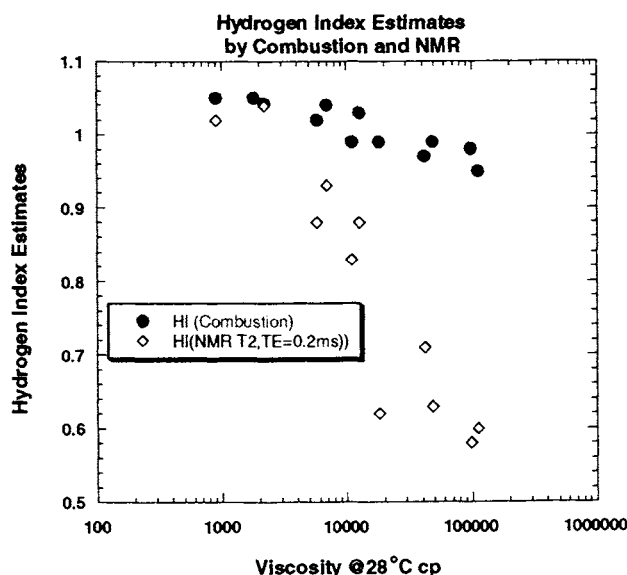


Figure 3.3: Hydrogen Index (HI) determinations on high density oils using combustion and low field NMR methods. (From LaTorraca et al. [135].)

The HI estimates at a Larmor frequency of 2 MHz are plotted in Figure 3.4 for various echo spacings. The apparent HI values depend strongly on the echo spacing. This dependence was explained as being caused by the fast T_2 relaxation rate of high density crude oils being too rapid to be adequately sampled by the low field NMR instruments. This resulted in the low values of apparent HI shown for all the echo spacings used.

The foregoing discussion on HI is mostly for dead oils. For live oils, the dissolved gas can significantly reduce the density such that the HI can be much less than one. Figure 3.5 shows a result for live oils where the HI is between 0.8 and 0.9. The linear lines are HI for different hydrogen:carbon ratios. The curve of G&P is the correlation developed by Gaymard and Poupon [92] based on alkanes as a function of density. It will not describe the HI for heavy oils because heavy oils contain a significant aromatic and polyaromatic content. The curve of K&V is the correlation of HI and the API gravity by Kleinberg and Vinegar [128].

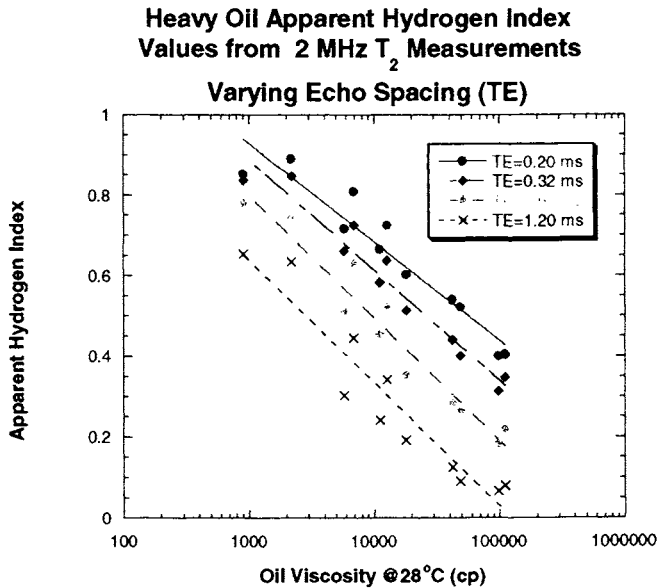


Figure 3.4: Low field NMR HI estimates vs viscosity with varying echo spacing (TE). (From LaTorraca et al. [135].)

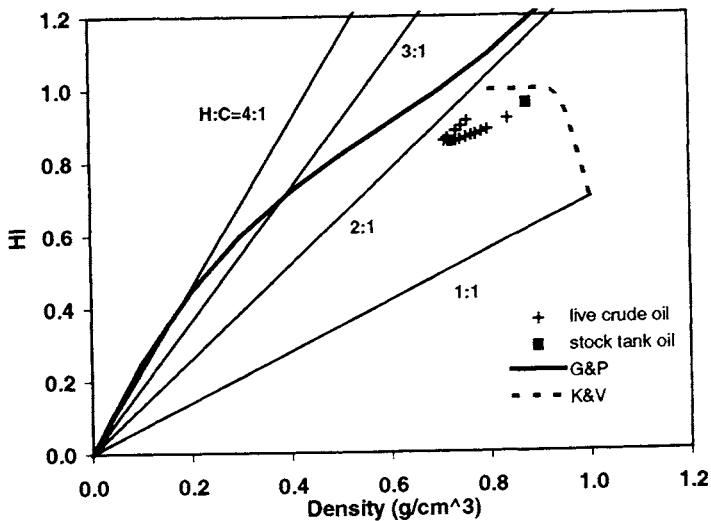


Figure 3.5: HI of live crude oils calculated from PVT data and measurements at ambient condition. (From Zhang et al. [230].)

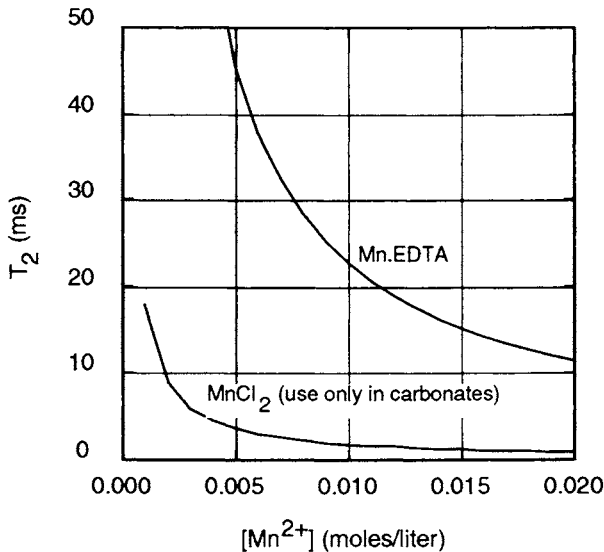


Figure 3.6: The T_2 relaxation time versus the concentrations of Mn^{++} for MnEDTA and $MnCl_2$. (From Horkowitz et al. [103].)

3.2 NMR properties of bulk fluids

3.2.1 Water

At room temperatures and at the low magnetic fields used in NMR well logging, the T_1 and T_2 values of bulk water are in the range of 1 to 3 seconds. The presence of dissolved oxygen, which is paramagnetic, accounts for much of the variation seen in T_1 and T_2 at low magnet strengths. Ferromagnetic and/or paramagnetic ions (such as iron or manganese) can greatly reduce the T_1 and T_2 values of brine. The dependence of T_2 on the concentrations of MnEDTA and $MnCl_2$ is shown in Figure 3.6, and its dependence on the temperature is shown in Figure 3.7. Note that only small concentrations of Mn are needed to drastically reduce the T_2 value.

Because the bulk T_1 and T_2 values for water are so long, they usually contribute very little to the measured relaxation rate of brine saturated porous media. The measured relaxation rate of rocks is normally dominated by surface relaxation which is covered in Chapter 4. The method of doping the borehole fluids or the drilling mud with ferro- or paramagnetic ions, which drastically reduce their T_1 and T_2 , is frequently used for residual oil determi-

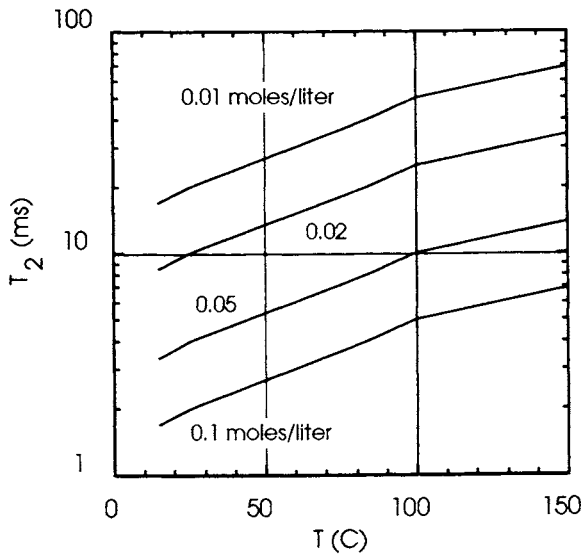


Figure 3.7: The T_2 relaxation time versus temperature for various concentrations of MnEDTA. (From Oakes and Smith [172].)

nation. This is discussed in Chapter 5.

3.2.2 Crude oils

Oil is often the non-wetting phase. It resides in the center of the pore space, and is prevented from making contact with the pore wall. In this situation, there is only bulk relaxation, which is unaffected by the formation and is only a function of oil composition and formation temperature.

However, if the oil wets the surface, the surface relaxation mechanism comes into play just in the same manner as it does for water. The shift of the oil relaxation times when the oil goes from a non-wetting phase to a partially wetting phase can be used to study the wettability issue. This is discussed in Chapter 4. For now, we shall assume that oil is the non-wetting phase, and its NMR relaxation time is the sole property of the oil itself.

For pure alkanes, pure oils, or light crudes, the T_1 or T_2 has a single, well-characterized value. However, for crude oils in general, they are usually characterized by a distribution of relaxation times. In this case, the T_1 or T_2 is represented by multiple amplitudes f_j corresponding to a distribution of relaxation times T_j . There are more discussions on this in Chapters 4 and 6.

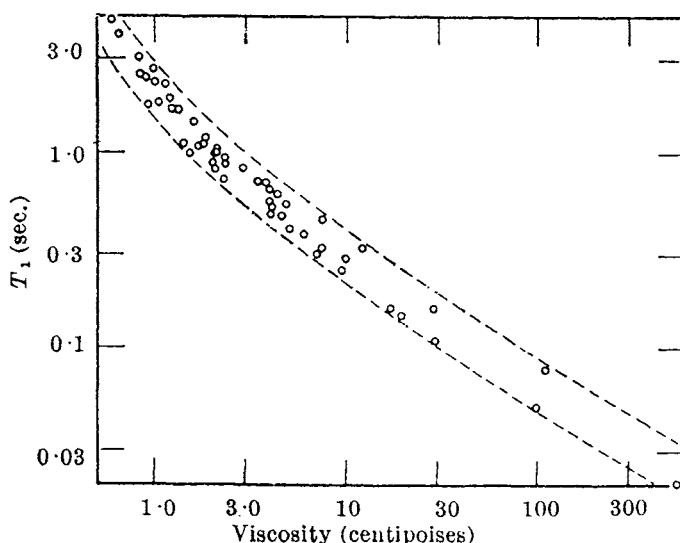


Figure 3.8: T_1 relaxation times versus viscosity for 14 crude oils at various temperatures. (From Brown [40].)

In 1961, Brown [40] studied the T_1 relaxation times for a number of crudes over a wide range of temperatures and viscosities. The relaxation times are obtained from the reciprocals of the initial slopes of the relaxation curves, that is, the reciprocals of the average relaxation rates. His results are shown in Figure 3.8.

Later, Morriss et al. [169] measured the T_2 distributions of a variety of crude oil samples in a uniform magnetic field (Figure 3.9). They found that, in contrast with refined oils which have narrow T_2 distributions, these crude oils have T_2 relaxation times which span several decades.

As shown in Figure 3.9, when the viscosity increases from 2.7 cp at the top of the left hand panel to 4304 cp at the bottom of the right hand panel, the T_2 distribution starts to broaden with the increase of the short relaxation time component, and eventually loses all the long time components and retains only the short time components when the oil is very viscous. This reflects the nature of the crudes which are mixtures of various hydrocarbon types, the mobile ones with small molecular sizes as well as the less mobile ones with long hydrocarbon chains.

Frequently, a single value, such as the logarithmic mean of the T_2 distribu-

tion, is used to characterize the overall property of T_2 . If the T_2 distribution has a set of multiple relative amplitudes f_j corresponding to a distribution of relaxation times T_j , then the logarithmic mean of the relaxation times, $T_{2,LM}$ is defined as:

$$\log T_{2,LM} = \frac{\sum_j f_j \log T_j}{\sum_j f_j}. \quad (3.4)$$

This logarithmic mean of relaxation times, $T_{2,LM}$, is sometimes referred to as the geometric mean, T_{2G} . Both $T_{2,LM}$ and viscosity for each sample are shown in Figure 3.9.

3.2.3 Gas

For most liquids and solids, the spins relax by dipole-dipole interaction. This relaxation is caused by the interaction of the magnetic dipoles of the nucleus of interest with the dipoles of other nuclei. The relaxation rate is proportional to η/T_K (i.e., the viscosity/absolute temperature). For simple gases, such as methane, the spins relax primarily by spin-rotation interaction. This is the coupling between the magnetic moments of the protons and the magnetic field produced at their positions by the rotation of the molecule. In contrast with fluids and solids, this kind of relaxation rate is proportional to T_K/η .

Unlike liquids, the T_1 of methane has an opposite behavior: It increases as the pressure increases, and decreases as the temperature increases. Figure 3.10 shows the T_1 of methane gas versus temperature and pressure.

A simple power law was found to describe the data shown in Figure 3.10 quite well [93, 182, 178]:

$$T_{1,g} = 2.5 \times 10^4 \frac{\rho}{T_K^{1.17}}. \quad (3.5)$$

where T_K is the absolute temperature in Kelvin, and ρ is the density in g/cm^3 .

3.2.4 Mud filtrate

Water based mud

There are various kinds of water based drilling muds. It is always advisable to measure the T_1 and T_2 of the mud filtrate before a logging job. Depending on the degree of invasion of the mud filtrate and the depth of the probed

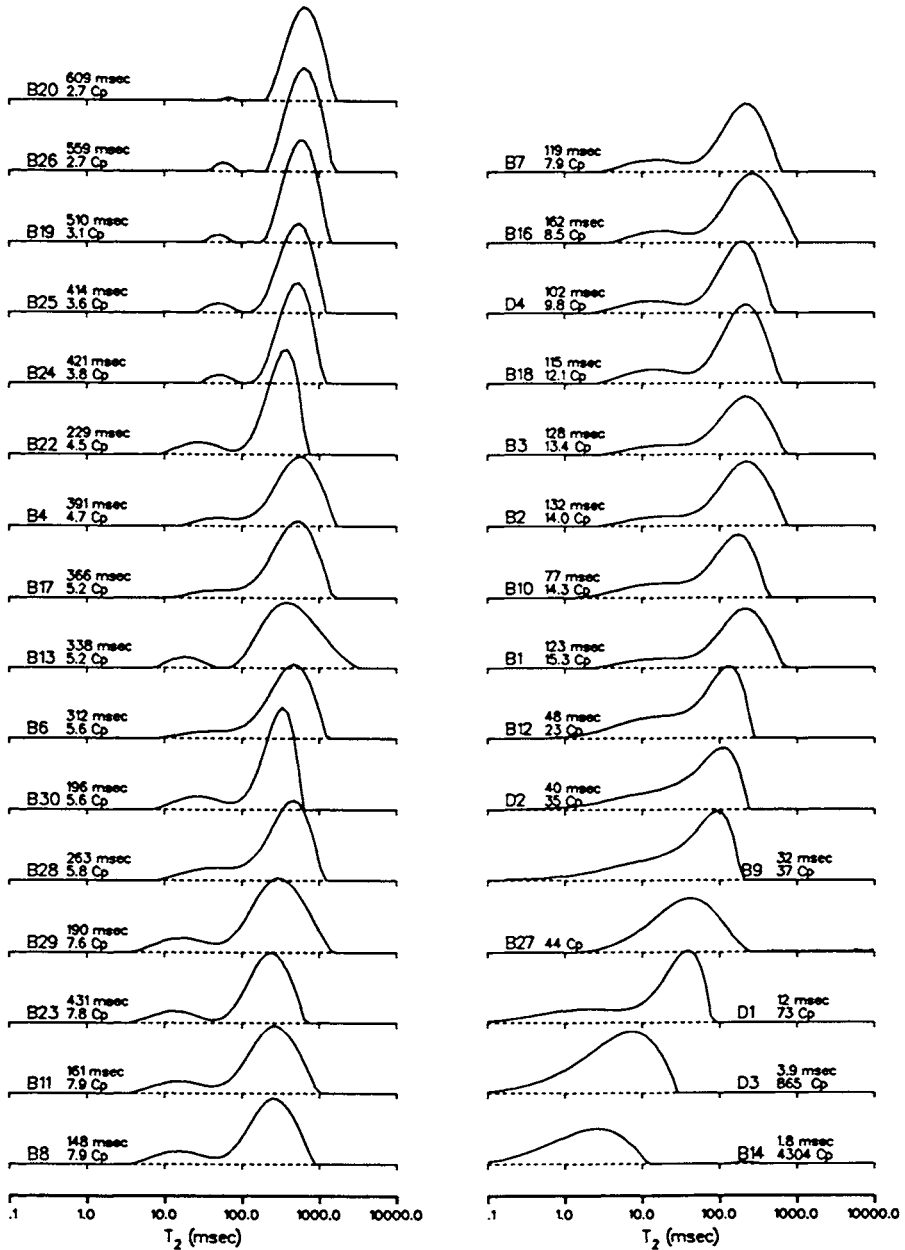


Figure 3.9: The T_2 distributions for bulk oil samples from Belridge field plotted in order of increasing viscosity. The sample number, logarithmic T_2 mean, $T_{2,LM}$, and measured viscosity (in centipoise) are shown for each sample. (From Morriss et al. [169].)

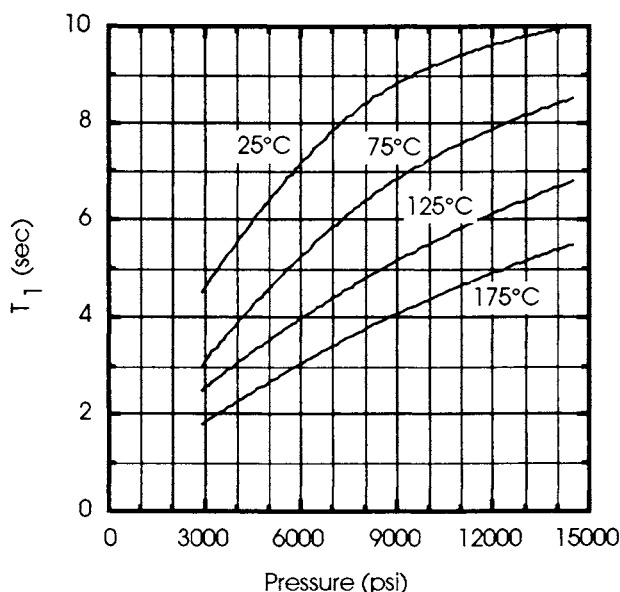


Figure 3.10: T_1 of methane gas versus temperature and pressure. (From Gerritsma et al. [93].)

zone of the logging tools, frequently we are measuring the mud filtrate saturated portion of the formation. In some water-based drilling muds, additives can reduce the relaxation time significantly. An example of such a mud is chromium or ferro-chromium lignosulfonate mud. Figure 3.11 shows T_1 as a function of thinners added to drilling mud from the early work of Coolidge et al. [66].

Oil based mud

Oil based drilling mud (OBM) is used both for environmental reasons and to ensure the stability of the borehole. Oil-based muds typically contain a base oil and at least two additives such as a surfactant and an emulsifier. The purpose of the surfactant is to make the solid particles in the mud oil-wet to block water that is not emulsified from entering the formation. The purpose of the emulsifier is to maintain the water in the mud in an emulsified state. The filtrate from these muds can invade the formation and displace and/or mix with the reservoir fluids. There is no known published study yet on how the various OBM filtrates interact with reservoir fluids and affect the T_2

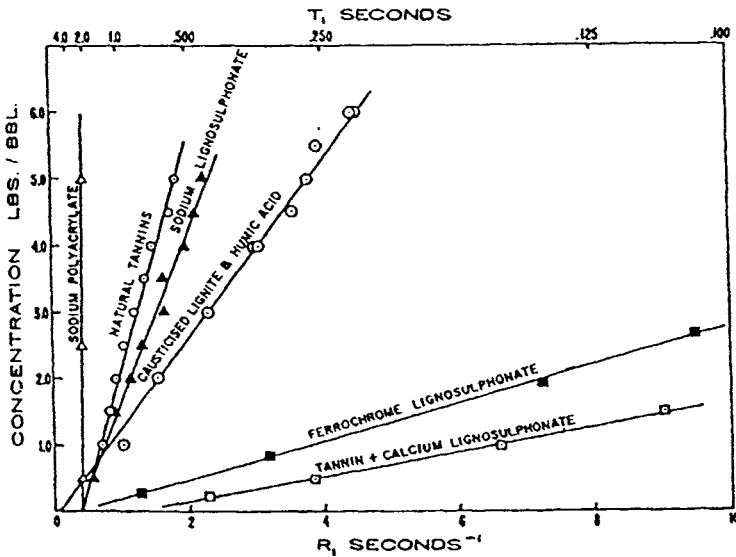


Figure 3.11: Concentration of thinners added to drilling mud versus T_1 relaxation rate, $R_1 \equiv 1/T_1$. (From Coolidge et al. [66].)

measured by NMR logging tools. However, it is known that the presence of these additives in the OBM filtrate can affect the T_2 response.

Figure 3.12 shows the results of the T_2 measurements on the fluid constituents of an oil based mud plus a filtrate sample from the mud. The fluid constituents were the base oil, emulsifier, and surfactant. Note that the T_2 distribution for the base oil is essentially a single component while the surfactant and the emulsifier have complicated, bimodal distributions. Note also that the T_2 for the filtrate is shorter than the base oil T_2 and that a shorter peak indicates that either paramagnetic impurities or some of the surfactant or emulsifier is in the filtrate. Additionally, the surfactant has very short components consistent with its asphaltene (high viscosity, short T_2) constituent.

3.3 T_2 in a magnetic field gradient

For T_2 relaxation in a magnetic field gradient, the dephasing of spins in an infinite medium due to diffusion results in additional signal decay. This leads

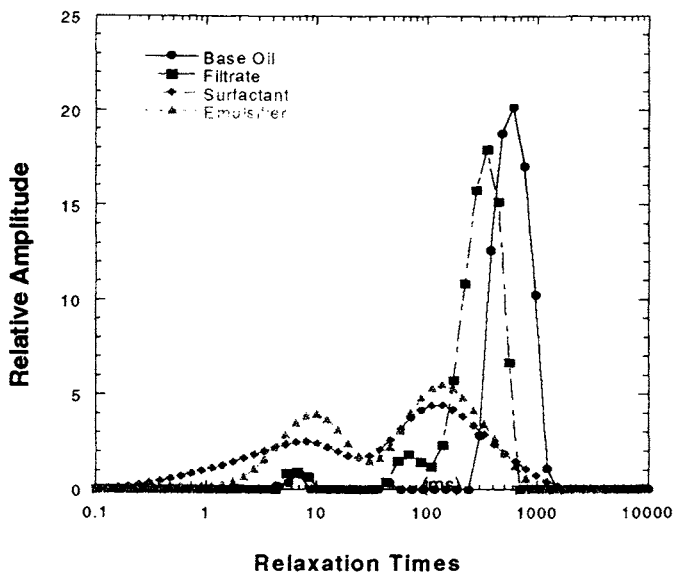


Figure 3.12: T_2 distributions of an oil based mud (containing base oil, surfactant, and emulsifier) and the mud filtrate. (From LaTorraca et al. [136].)

to an additional effective relaxation rate $1/T_{2D}$

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \frac{1}{T_{2D}}, \quad (3.6)$$

where

$$\frac{1}{T_{2D}} \equiv \frac{(\gamma G t_E)^2 D}{12} \quad (3.7)$$

and γ is the gyromagnetic ratio, G the magnetic field gradient (assumed to be constant), t_E (TE) the time between echoes, D the self diffusion coefficient of the fluid, and T_{2B} the bulk relaxation time (see Chapter 2 for additional details).

Diffusion of spins in a magnetic field can cause incomplete refocusing in a spin echo experiment. This leads to a reduction in signal amplitude in addition to the regular T_2 decay. Since this is an issue of dephasing of spins due to diffusion, it is strictly a T_2 effect (non-existent in T_1)². The net result of this diffusion effect is equivalent to having an additional relaxation rate term as shown in Eqs. (3.6) and (3.7).

²See footnote [1] of Chapter 2 on p. 29.

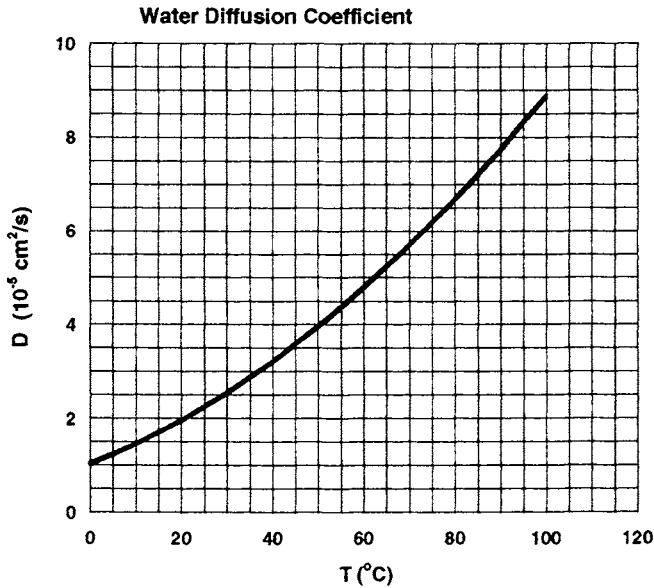


Figure 3.13: Diffusion coefficient of water as a function of temperature³.

When diffusion is restricted in porous media, the diffusion coefficient in Eq. (3.7) may have to be replaced by the restricted diffusion coefficient for a large TE. There are also other issues as to the validity of Eq. (3.7) which we shall discuss in detail in Chapter 7. For illustrative purposes, we shall use the free diffusion coefficient of the fluid to calculate T_{2D} and describe diffusion effects on the overall relaxation rate.

Looking at the expression of Eq. (3.7), the diffusion coefficient, the magnetic field gradient, and the time between echoes (TE), all play roles in determining T_{2D} .

Water

The diffusion coefficient of water is quite insensitive to pressure, but depends strongly on temperature, as shown in Figure 3.13³. Note that the net effect

³Note that the plot is based on $D = a(T_K/\eta)$, where a is determined by $D = 2.0 \times 10^{-5} \text{ cm}^2/\text{s}$ at 20°C and the viscosity $\eta = 1.002 \text{ cp}$. The temperature dependence of η is based on Sengers and Watson [197]. This D vs. T plot can be approximated by the empirical fit: $D = 1.0413 + 0.039828T + 0.00040318T^2$ for the range of 0° to 100°C , where T is in Celsius and D is in $10^{-5} \text{ cm}^2/\text{s}$.

of this diffusion term, $1/T_{2D}$, is to shift the T_2 relaxation time at zero field gradient (intrinsic relaxation time) to a shorter apparent relaxation time. Figure 3.14 shows such an effect computed for water using the following input

$$\begin{aligned}\gamma &= 2.6752 \times 10^4 \text{ rad sec}^{-1} \text{gauss}^{-1} \\ D &= 2.5 \times 10^{-5} \text{ cm}^2/\text{sec} \\ \text{TE} &= 1.2 \text{ and } 3.6 \text{ msec}\end{aligned}$$

Note that for a large intrinsic T_2 , the change in apparent T_2 value is greater, as illustrated by the bending of the curve, and the effect is more significant for larger field gradient values. For an intrinsic T_2 of 1 sec, at a TE of 3.6 ms, its apparent T_2 is about 340 ms at 10 Gauss/cm and about 115 ms at 20 Gauss/cm; whereas for an intrinsic T_2 of 10 ms, there is hardly any change in its apparent T_2 at the same field gradients.

Oil

Typically, the diffusion coefficient for oil is much less than that of water. Figure 3.15 shows the apparent T_2 for a range of diffusion coefficient values from $D = 1 \times 10^{-6}$ to $2.5 \times 10^{-5} \text{ cm}^2/\text{sec}$ in a field gradient of 20 Gauss/cm and a TE of 3.6 ms. Most oils have a diffusion coefficient near that of the top curve, whereas water has a diffusion coefficient near that of the bottom curve. Thus, there is very little shift of relaxation time for oils due to the field gradient, whereas the shift for water is quite significant. This is an important property which we use to distinguish oil from water in NMR logs.

Gas

The diffusion coefficient for gas, on the other hand, is much larger than that of water. The bulk (unrestricted) diffusion coefficient of methane is given by the following [173, 99]:

$$D = 8.5 \times 10^{-7} \frac{T_K^{0.9}}{\rho} \quad (3.8)$$

where T_K is the absolute temperature in Kelvin, and ρ is the density in g/cm^3 . The density of methane versus temperature is shown in Figure 3.16 [8] and its diffusion coefficient versus temperature is shown in Figure 3.17.

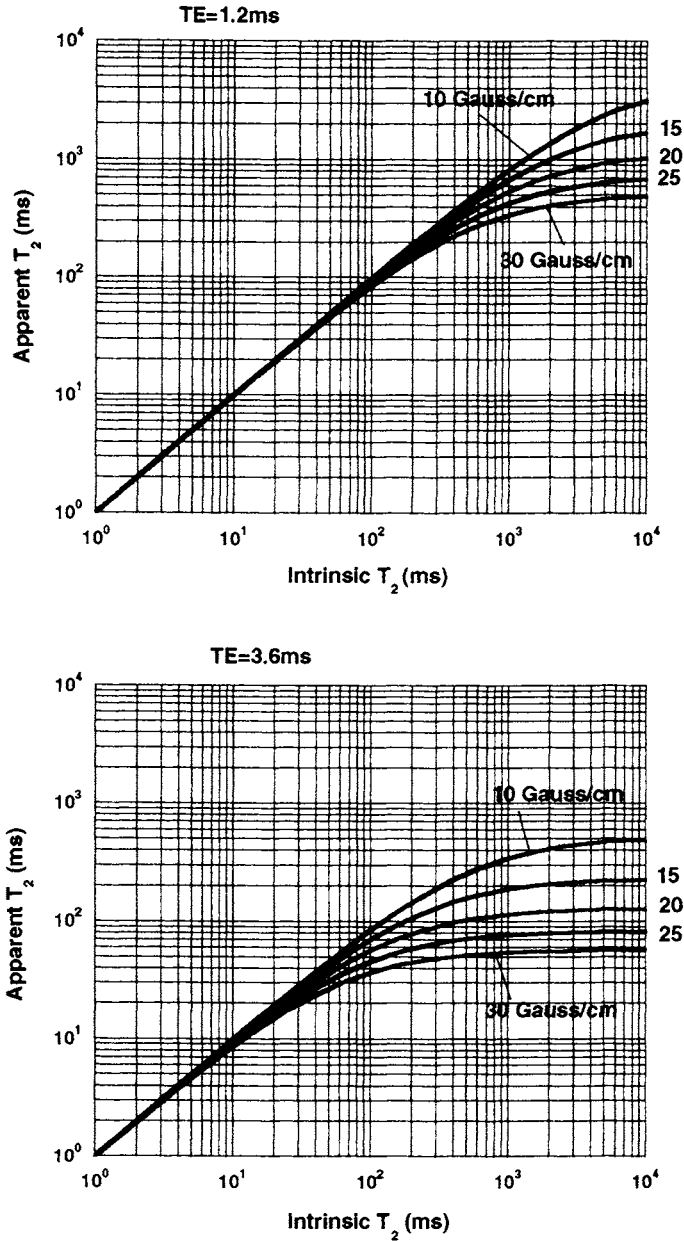


Figure 3.14: The apparent T_2 relaxation time versus intrinsic T_2 relaxation time in a constant magnetic field gradient of various strengths in Gauss/cm and a TE of 1.2 (top) or 3.6 (bottom) ms.

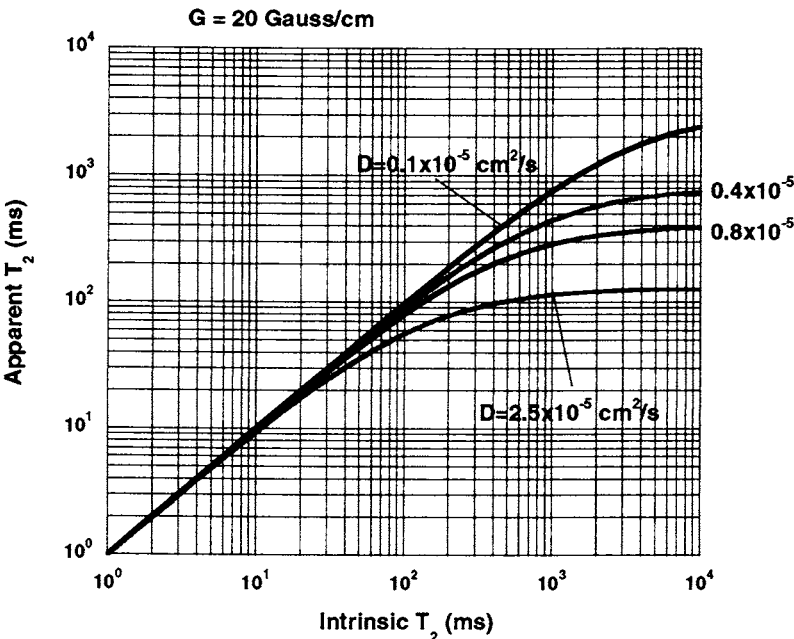


Figure 3.15: The apparent T_2 relaxation time versus intrinsic T_2 relaxation time for four values of diffusion coefficient in a constant magnetic field gradient of 20 Gauss/cm and a TE of 3.6 ms.

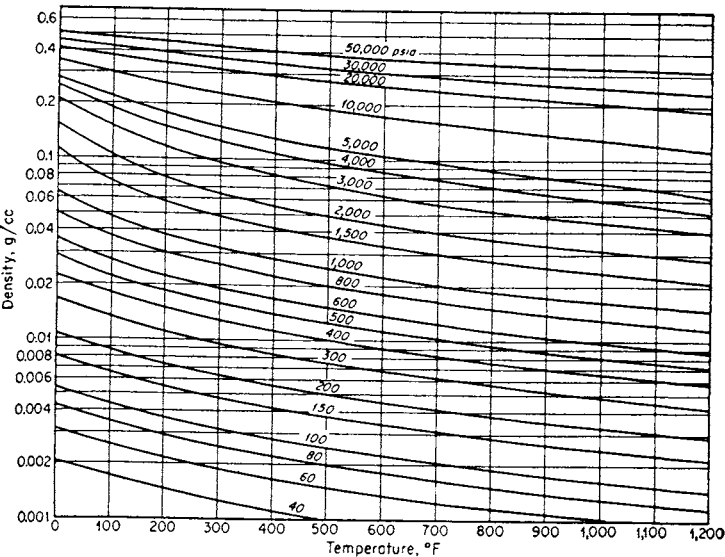


Figure 3.16: The density of methane versus temperature at various pressures. (Reproduced from Fig. 4.17 of Ref. [8] with permission of McGraw-Hill.)

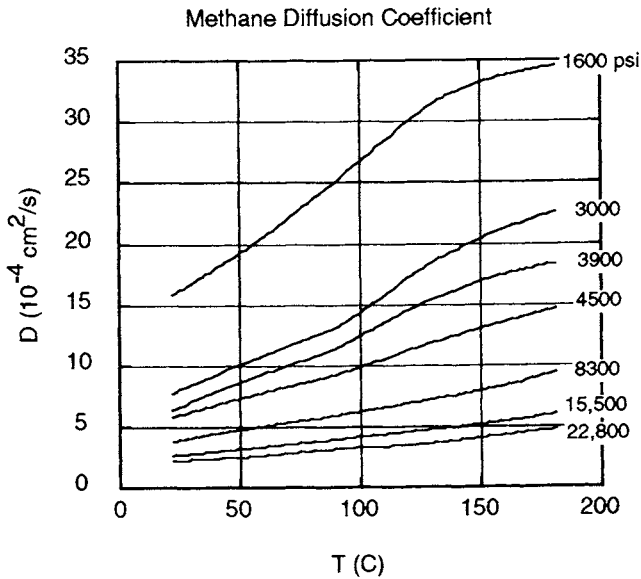


Figure 3.17: The diffusion coefficient of methane versus temperature at various pressures. (From Kleinberg and Vinegar [128].)

A typical value for the diffusion coefficient for methane is of the order of $10^{-3} \text{ cm}^2/\text{s}$. Since T_{2B} is of the order of seconds, a value of D of the order of $10^{-4} \text{ cm}^2/\text{s}$ already makes $1/T_{2D}$ the dominant term on the RHS of Eq. (3.6). Thus, for gas, we have

$$T_2 \approx T_{2D} \equiv \frac{12}{(\gamma G t_E)^2 D}. \quad (3.9)$$

Figure 3.18 shows the apparent T_2 for gas versus its diffusion coefficient in a constant field gradient of 20 Gauss/cm at different TEs. In generating Figure 3.18, we have assumed a T_{2B} for methane of 6 s and used Eq. (3.6) for our calculation. Such an assumption is irrelevant for a TE of 1.2 or 3.6 ms, as the approximation made in Eq. (3.9) is reasonably valid, whereas for a TE of 0.2 ms, a full calculation using Eq. (3.6) is needed.

3.4 Oil viscosity and diffusion constant

One of the important NMR properties of oils is the correlation of the T_1 or T_2 with the viscosity and diffusion coefficient of the oils. This is an important

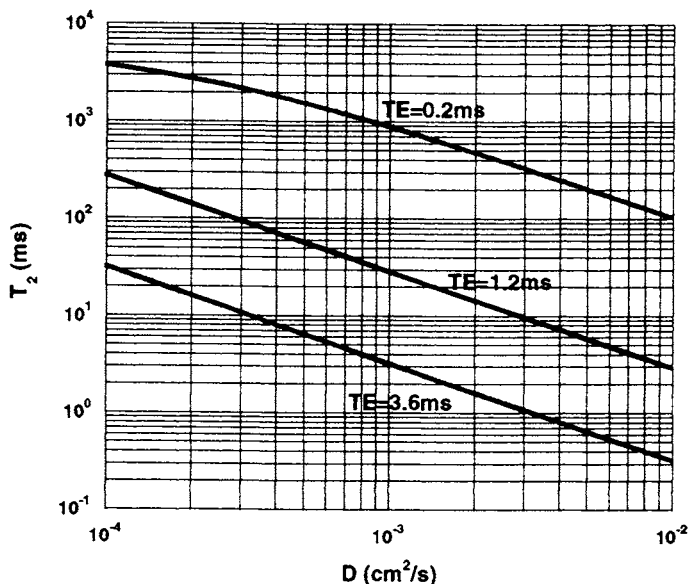


Figure 3.18: Calculated apparent T_2 of methane vs. D for unrestricted diffusion in a field gradient of 20 Gauss/cm.

correlation which is used for the estimation of oil viscosity from the T_2 values obtained from the NMR logs.

3.4.1 Viscosity correlation

Early measurements of T_2 relaxation times of dead crude oils by Vinegar [220] and Morris et al.[169] (Figure 3.19) have established a linear relationship between the logarithmic mean of T_2 and viscosity on a log-log plot

$$T_{2,LM} = \frac{1.2T_K}{298\eta} = 0.00403 \frac{T_K}{\eta} \quad (\text{contains dissolved oxygen}) \quad (3.10)$$

where $T_{2,LM}$ is in sec, T_K is the absolute temperature in degree Kelvin, and η is the viscosity in centipoise. The above measurements were done without the deoxygenation procedure, and the samples presumably contained dissolved oxygen of varying degree. The latter is paramagnetic and is known to cause increased relaxation, i.e., shortening of the relaxation time.

Later work by LaTorraca et al. [134, 135] on heavy oils at 2 MHz has shown that the apparent $T_{2,LM}$ may depend on echo spacing. For heavy oils

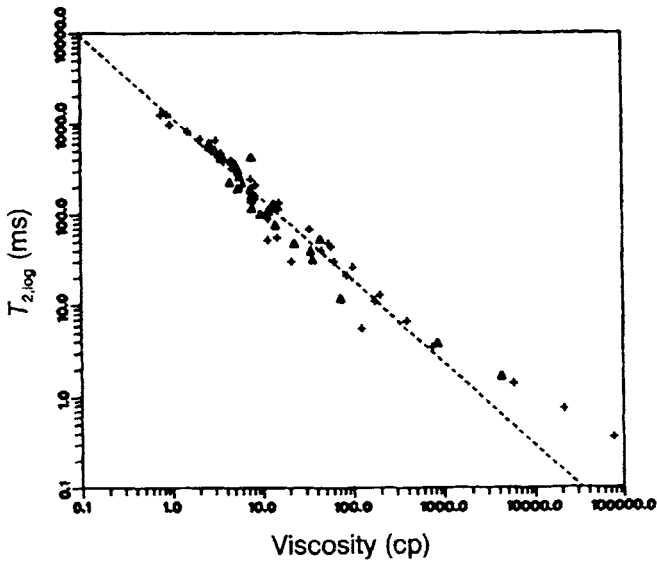


Figure 3.19: $T_{2,LM}$ versus crude oils. The best fit shown in Figure is $T_{2,LM} = 1200/\eta^{0.9}$ where $T_{2,LM}$ is in ms. Later, a more general form as Eq. (3.10) is adopted. (From Morriss et al. [169].)

at large TE, the loss of signal may result in a higher apparent $T_{2,LM}$. They established a formula for correcting this effect:

$$\eta = \left(\frac{2210 + 469 t_E^2}{T_{2,LM} - (t_E + 0.5)} \right) \left(\frac{T_K}{298} \right) \quad (3.11)$$

where t_E (TE) and $T_{2,LM}$ are in ms. Figures 3.20 and 3.21 show the results of their experiments.

Other measurements by Zega et al. [229] and Zhang et al. [230] on deoxygenated pure alkane and alkane mixtures also established a linear relationship between $T_{1,2}$ and viscosity on a log-log plot⁴

$$T_{1,2} = 0.00713 \frac{T_K}{\eta} \quad (\text{oxygen free}), \quad (3.12)$$

and later work by Lo et al. [143, 144] has revised the fitting constant:

$$T_{1,2} = 0.009558 \frac{T_K}{\eta} \quad (\text{oxygen free}), \quad (3.13)$$

⁴Note that Eqs. (3.10), (3.12), and (3.13) are all based upon room temperature measurements. Thus, the T_K dependence is based on theory.

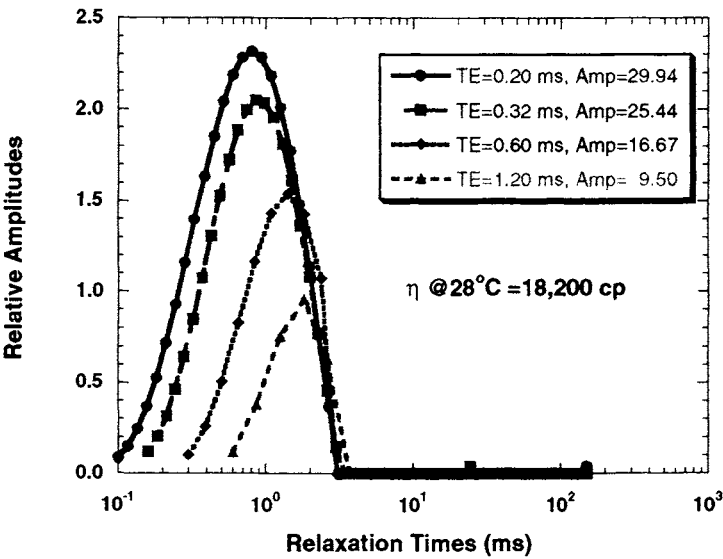


Figure 3.20: T_2 distributions of a heavy oil with varying echo spacing measured at 2 MHz. Note reduction of amplitude and apparent shift of time constants to longer times with increasing TE. (From LaTorraca et al. [135].)

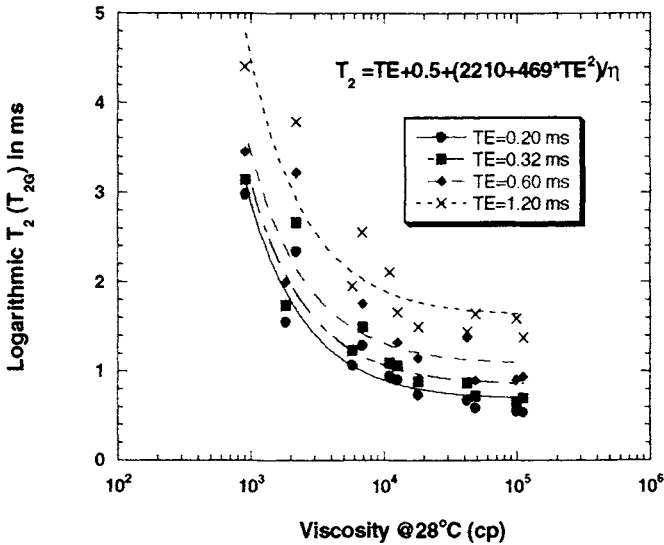


Figure 3.21: Logarithmic mean of T_2 of heavy oils versus viscosity at 2 MHz for varying TE. (From LaTorraca et al. [135].)

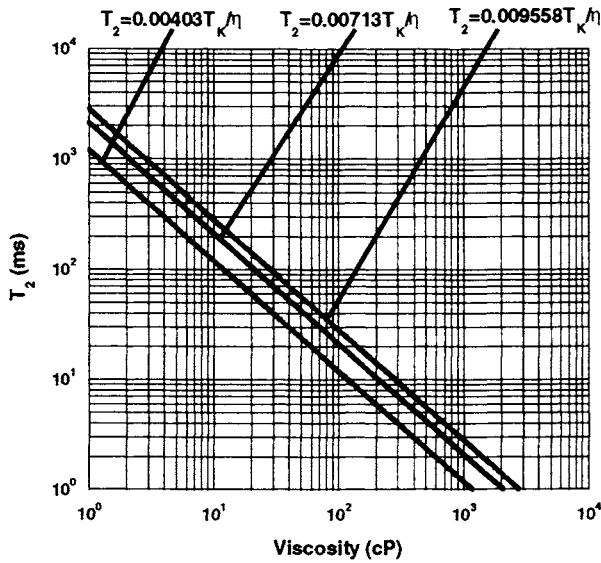


Figure 3.22: $T_{1,2}$ versus viscosity for Eqs. (3.10), (3.12), and (3.13) at room temperature.

where $T_{1,2}$ is in seconds, T_K is the absolute temperature in degree Kelvin, and η is the viscosity in centipoise. The deoxygenation procedure yields a larger $T_{1,2}$ relaxation time, consistent with the hypothesis that the earlier measurements [i.e., Eq. (3.10)] may have dissolved oxygen in the samples. The relationships between $T_{1,2}$ and viscosity of Eqs. (3.10), (3.12), and (3.13) are plotted in Figure 3.22.

3.4.2 Diffusion constants of oils

Using the Pulsed Field Gradient Spin Echo (PFGSE) method (Figure 2.9), Lo et al. [143] measured the diffusion coefficient of a liquid methane-decane mixture. The measured PFGSE amplitude is expressed as follows:

$$M(2\tau)/M(0) = \sum_i f_i e^{-\gamma^2 \delta^2 g^2 (\Delta - \frac{1}{3}\delta) D_i} \quad (3.14)$$

where the values of Δ (diffusion time), g (pulsed field gradient), and δ (pulse width) can be varied. They assumed a distribution of diffusion coefficients D_i and $\sum_i f_i = 1$, and obtained a distribution of self diffusion coefficients with

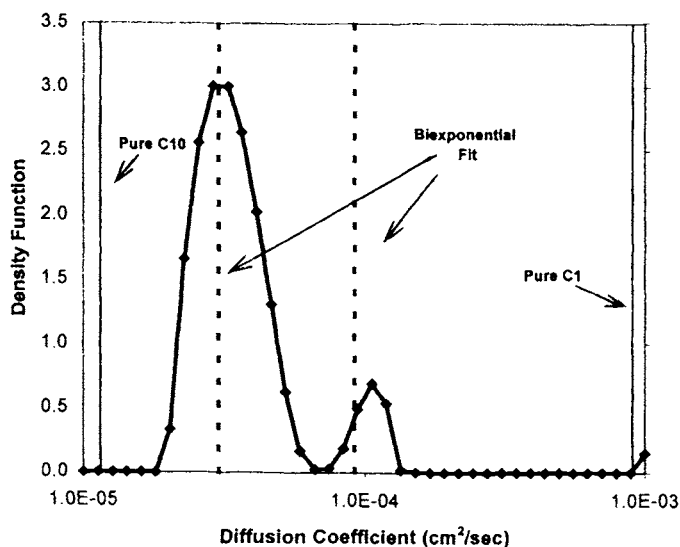


Figure 3.23: Diffusion coefficient distribution of a liquid methane-decane mixture. The proton fraction of methane is 0.187, and that of decane is 0.813; whereas the relative amplitudes in the distribution are 0.189 and 0.811, respectively, quite consistent with the proton fractions. (From Lo et al. [143].)

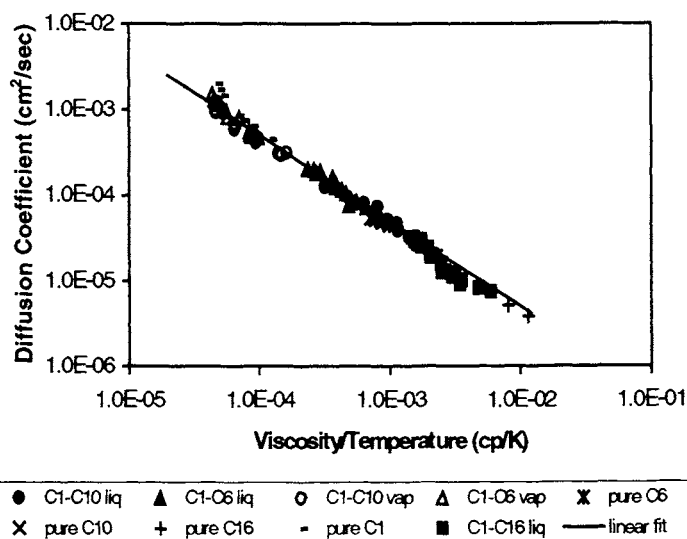


Figure 3.24: Diffusion coefficient dependence on viscosity/temperature for pure methane, pure alkanes, and methane-alkane mixtures. The dependence is linear regardless of composition. (From Lo et al. [144].)

varying relative amplitudes. Their results (shown in Figure 3.23) indicate that there are two components in the diffusion coefficient distribution, each is different from their values in the pure state, while their amplitudes are quite consistent with the relative mole fractions of the constituents.

Further work by Lo et al. [144] shows that the logarithmic mean of the diffusion coefficient distribution of a number of methane-alkane mixtures varies linearly with viscosity/temperature on a log-log plot, regardless of composition. Their results are shown in Figure 3.24. The relation they found can be described by the following equation:

$$D = 5.05 \times 10^{-8} \frac{T_K}{\eta} \quad (3.15)$$

where D is in cm^2/sec , T_K is the absolute temperature in degree Kelvin, and η is the viscosity in centipoise (cp).

3.4.3 Correction for live oils

The linear relationships on a log-log plot between $T_{2,LM}$ and η/T_K , as discussed earlier, are for dead oils. Lo et al. [144] found that the relationship for live oils is no longer linear. Based on experimental measurements on the T_1 relaxation time for pure methane, pure alkanes, and three methane-alkane mixtures (methane-hexane, methane-decane, and methane-hexadecane), they found the T_1 dependence on viscosity/temperature departs from linear correlation, as shown in Figure 3.25.

The T_1 dependence on diffusion coefficient for the systems they studied is plotted in Figure 3.26 for various gas oil ratios (GOR). The GOR is defined as:

$$\text{GOR} = \frac{\text{cubic meters of methane at standard condition}}{\text{cubic meters of stock tank oil}} \quad (3.16)$$

where the standard condition is 60°F and 1 atmospheric pressure.

Since T_1 versus T_K/η and T_1 versus D share the same GORs, the horizontal axis can be either D in cm^2/sec or $5.05 \times 10^{-8} T_K/\eta$.

Figure 3.27 shows the deviation from the linear relationship in the log scale, which is plotted vs. GOR, and a quadratic functional fit determined as follows:

$$\begin{aligned} \text{deviation} &= \log_{10} T_{1,linear} - \log_{10} T_1 = \log_{10}(f(\text{GOR})) \\ &= -0.127(\log_{10} \text{GOR})^2 + 1.25 \log_{10} \text{GOR} - 2.80 \end{aligned} \quad (3.17)$$

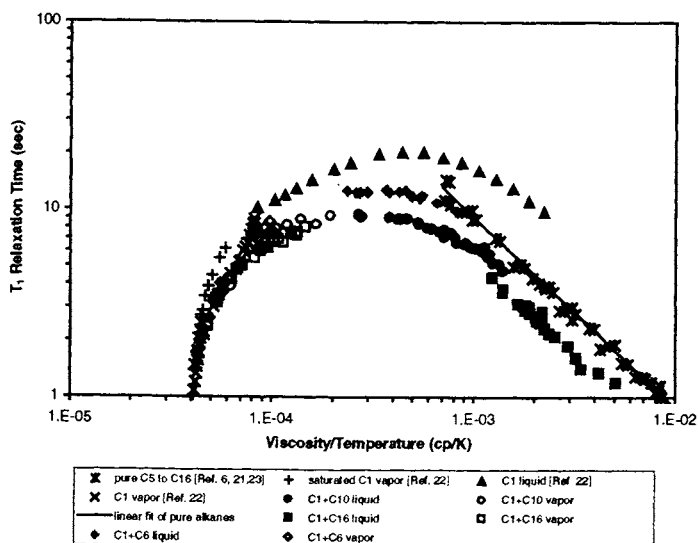


Figure 3.25: T_1 relaxation time for pure methane, pure alkanes, and three methane-alkane mixtures. Its dependence on viscosity/temperature departs from linear correlation. (From Lo et al. [144].)

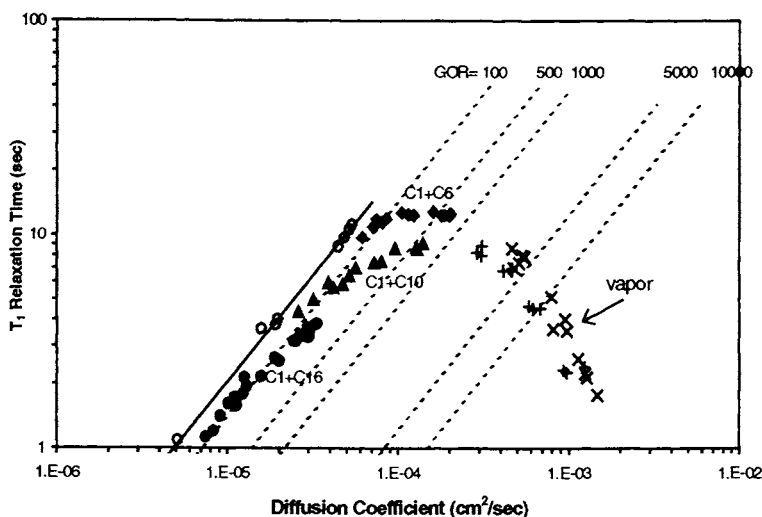


Figure 3.26: T_1 dependence on diffusion coefficient for pure methane, pure alkanes, and methane-alkane mixtures. Contour lines of constant GOR are plotted parallel to zero GOR. (From Lo et al. [144].)

where $T_{1,linear}$ is given by the earlier equations. So, once the GOR and the T_1 of the oil are known, the viscosity can be estimated as follows:

$$T_1 = \frac{0.009558}{f(\text{GOR})} \left(\frac{T_K}{\eta} \right) \quad (3.18)$$

where Eq. (3.13) is used for the linear relation between T_1 and T_K/η .

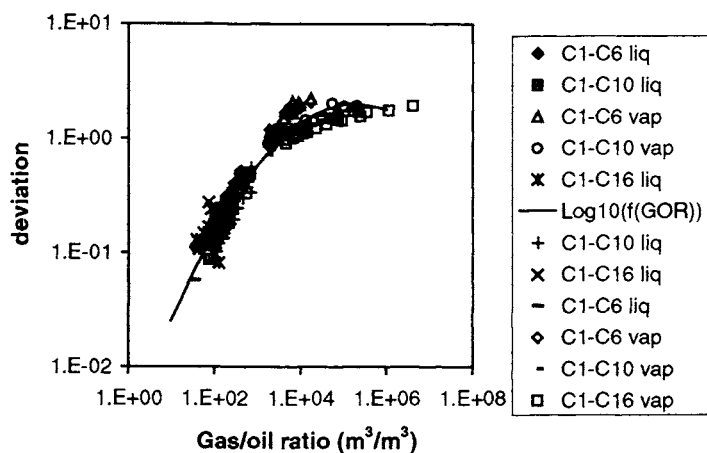
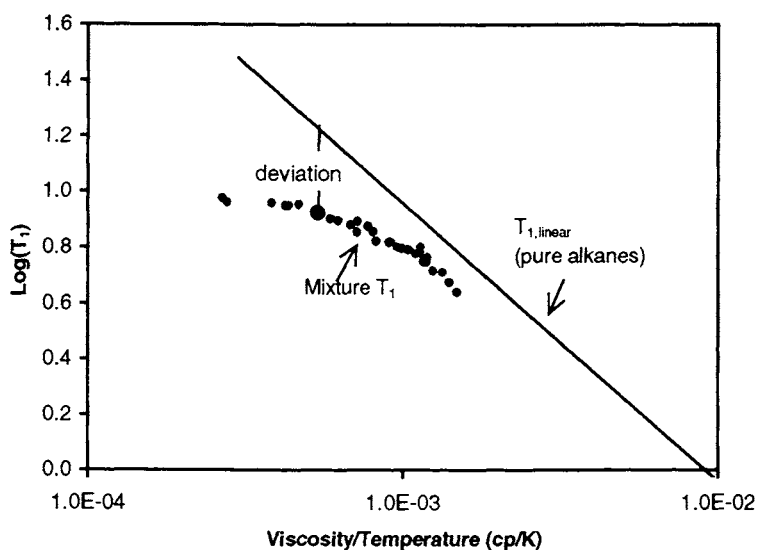


Figure 3.27: The deviation from the linear relationship in the log scale (top) was plotted against GOR (bottom), and a quadratic functional fit was determined and used for correction for live oils. (From Lo et al. [144].)